Exponential growth and decay

1 For each of the following exponential functions, where x represents time:

i Classify the function as exponential growth or exponential decay.

ii Find the rate of growth or decay per time period as a percentage.

a $y = 0.68\,(1.6)^x$ **b** $y = 0.98\,(0.2)^x$ **c** $y = 100\,(2.05)^x$ **d** $y = 50\,(0.78)^x$

2 The mass in kilograms, M, of a baby orangutan at n months of age is given by the equation $M = 1.8 \times 1.1^n$, for ages up to $n = 6$ months.

a Find the mass, M, of a baby orangutan at 3 months of age to one decimal place.

b Complete the table, rounding your answers to one decimal place:

Months (n)	0	1	2	3	4	5	6
Mass (M)							

c Sketch a graph of the mass function.

3 The frequency f (Hz) of the nth key of an 88-key piano is given by $f\,(n) = 440\left(2^{\frac{1}{12}}\right)^{n-49}$.

a Determine the frequency of the forty-ninth key.

b Determine the frequency of the 40th key. Round your answer to the nearest whole number.

c Find the value of n that corresponds to the key with a frequency of 1760 Hz.

4 Marlow purchased a piece of sports memorabilia for $\$1300$, and it is expected to increase in value by 8% per year.

a Write a function, y, to represent the value of the piece of sports memorabilia after v years.

b Find the value of the piece of sports memorabilia after 10 years.

5 Due to favorable market conditions, a company's annual costs are expected to decrease by 5% per year for the next few years. Their annual costs for the current financial year are $\$322\,000$.

a Write a function, y, to represent the company's annual costs in m years.

b Find the company's annual costs in 5 years.

6 A ball dropped from a height of 21 yd will bounce back off the ground to 50% of the height of the previous bounce (or the height from which it is dropped when considering the first bounce).

a Write a function, y, to represent the height of the nth bounce.

b Calculate the height of the fifth bounce, correct to two decimal places.

7 A sample contains 300 grams of carbon-11, which has a half-life of 20 minutes.

a Write a function, A, to represent the amount of carbon-11 remaining in the sample after n minutes.

b Find the amount of the isotope left after after 3 hours. Round your answer to two decimal places.

8 The area covered by an ice shelf was measured over some warmer months. At the beginning of the first month the ice shelf was spread over an area of 1437 km^2, and it was found that this area decreased by 2% each month over the recording period.

a Find the area that was covered by the ice shelf after 13 months of recording. Round your answer to the nearest square kilometer.

b Find the loss in the ice shelf's area over 13 months.

9 A new appliance is valued at $\$990$. Each year it is worth 9% less than the previous year's value.

a Find the value of the appliance after the first year.

b Find the value of the appliance after the second year.

c Find the equation that relates the value of the appliance, A, with the number of years passed, t.

d Find the value of the appliance after twelve years.

10 A car originally valued at $\$28\,000$ is depreci[redacted]t the rate of 15% per year. The salvage value, S, of the car after n years is given by $S = 28\,000\left(1 - \frac{15}{100}\right)^n$.

a Complete the following table. Round your answers to two decimal places.

n	2	4	6	8	10	12
S						

b Sketch the graph of $S = 28\,000\left(1 - \frac{15}{100}\right)^n$.

c What is the value of the car after 3 years?

d After how many years does the value of the car drop down to an amount of $\$6\,485.27$?

11 A new car purchased for $\$38\,200$ depreciates at a rate $r\%$ each year. The value of the car for the first two years is shown in the table below:

years passed (n)	0	1	2
value of car $(\$A)$	$38\,200$	$37\,818$	$37\,439.82$

a Find the value of r.

b Write the rule for A, the value of the car n years after it is purchased.

c Assuming the rate of depreciation remains constant, how much can the car be sold for after 6 years?

d A new motorbike purchased for the same amount depreciates according to the model $V = 38\,200 \times 0.97^n$. Which vehicle depreciates more rapidly?

12 When a heated substance such as water starts to cool, its temperature T at time t minutes after being left to cool is given by an exponential function.

a Should T be an increasing or decreasing function?

b For positive values of a, state whether the following functions could be used to model the temperature T at time t:

i $T = -ab^t$, where $b > 1$.

ii $T = ab^{-t}$, where $b > 1$.

iii $T = ab^t$, where $0 \leq b \leq 1$.

iv $T = -ab^t$, where $0 \leq b \leq 1$.

c What does b represent in the model?

d Substance P starts off at a temperature of 145°C and is left to cool in a room whose temperature is a constant 0°C. By using the table below, find the equation for the temperature of the substance.

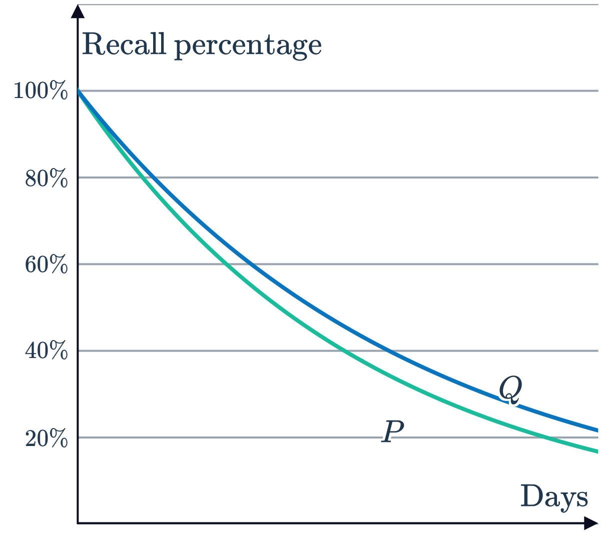
Minutes passed (t)	0	1	2	3	4
Temperature (T)	145	87	52.2	31.32	18.792

e Find the temperature after 10 minutes, correct to two decimal places.

f Will the temperature of the substance ever reach 0°C?

g The temperature of another substance Q which also starts off at a temperature of 145°C is modeled by $T = 145 \times 0.9^t$. Which substance will cool more rapidly?

13 In a memory study, subjects are asked to memorize some content and recall as much as they can remember each day thereafter. The given graphs represent the percentage of content remembered by two participants Maximilian (P) and Lucy (Q) after t days:



a Each day, Maximilian finds that he has forgotten 15% of what he could recount the day before. Form a model for the percentage P that Maximilian can still recall after t days.

b Who was forgetting more rapidly?

c Select the model for the percentage that Lucy can still recall after t days.

A $Q = (0.85)^t$ **B** $Q = (0.83)^t$

C $Q = (0.87)^t$ **D** $Q = (0.81)^t$

d On each day, what percentage of the previous day's content did Lucy forget?

e According to the models, will either of them completely forget the content?

Population growth

14 Switzerland's population in the next 10 years is expected to grow approximately according to the model $P = 8\,(1+r)^t$, where P represents the population (in millions) t years from now.
The world population in the next 10 years is expected to grow approximately according to the model $Q = 7130\,(1.0133)^t$, where Q represents the world population (in millions) t years from now.

a Use the model for P to find the current population in Switzerland.

b Use the model for Q to find:

i The current world population.

ii The population of the world in twenty years time, to the nearest million.

15 The population, P, of a particular town after n years is modeled by $P = P_0\,(1.6)^n$, where P_0 is the original population.
Find the population of the town after $3\frac{1}{2}$ years if its original population was $30\,000$. Round

16 The population, A, of aphids in a field of potato plants t weeks after intial observation is modeled by: $A = 1000 \times 2^t$.

a State the initial aphid population.

b Find the predicted aphid population after 5 weeks.

c Find the predicted aphid population 2 weeks before initial observations.

17 The population, P, of a town in millions after t years is approximated by the formula

$$P = 8\left(1 + \frac{2}{100}\right)^t$$

a Complete the table of values, rounding your answers to one decimal place:

t	5	10	15	20	25	30
P						

b Sketch the graph of the function $P = 8\left(1 + \dfrac{2}{100}\right)^t$.

c After how many years will the population reach 12.6 million?

18 The population of Joanne's home town is $27\,400$ and has a growth rate of 6% each year.

a Find the size of the population after 7 years.

b How many more people will be in the town after 10 years?

19 A country’s population of 18 million people is expected to increase by 2.1% each year over the next 38 years. Calculate the expected population in 38 years time.

20 Tina currently has 174 followers on her online business page. If the number of people following her is growing at a rate of 5% per month, how many people will be following her business page in 6 months?

21 A scientist observed that a certain species of bacteria grew by 13% each hour. How many bacteria will be present after 17 hours if there were 79 initially?

22 The population of some native bees is declining at a rate of 8% per year. If there are $10\,000$ bees in a hive now, how many will there be in 5 years?

23 The population of cows at a free range farm [redacted] growth rate of 17% p.a. If the farm originally has 140 cows, how many will there be after 4 years?

24 The population of two different bacteria, labeled J and K, are given by the tables below:
Bacteria J:

t (time in days)	0	1	2	3	4
P (population)	1	60	$3\,600$	2.16×10^5	1.296×10^7

Bacteria K:

t (time in days)	0	1	2	3	4
Q (population)	1	50	$2\,500$	1.25×10^5	6.25×10^6

a Does the population of bacteria J increase slower or faster than the population of K?

b The population P of bacteria J at time t can be modeled using the standard equation $P(t) = a^t$, where $a > 1$. By using the table of values, graph $P(t)$ for $t > 0$.

c The population Q of bacteria K at time t can be modeled using the standard equation $Q(t) = b^t$, where $b > 1$. By using the table of values, graph $Q(t)$ for $t > 0$.

d Can the population of bacteria K be found by dilating the population of bacteria J by a factor?

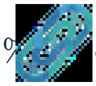
25 The population of a country is currently estimated at $2\,410\,000$ and is expected to grow by 2.2% each year. The country's food supply is currently enough for a population of $3\,374\,000$ people and is set to increase by 2% each year.

a Determine if there will ever be a food shortage. Explain your reasoning.

b Explain how you could use a graph to determine when a shortage may occur.

Compound interest

26 A fixed-rate investment generates a return of 6% per annum, compounded annually. The value of the investment is modeled by $A = P(1.06)^t$, where P is the original investment. Find the value of the investment after after $3\frac{1}{4}$ years if the original investment was $\$200$.

27 The value, A, of a $\$29\,000$ investment at 11%  compounded annually, after n years, can be modeled by: $A = 29\,000 \times 1.11^n$.

- **a** Calculate the value of the investment after 2 years.
- **b** Complete the table:

n	1	2	3	4	5	6
A						

- **c** Describe the rate of change of the investment.

28 The equation $A = 12\,000 \times 1.15^n$ can be used to calculate the value, A, of a $\$12\,000$ investment at 15% p.a., with interest compounded annually, after n years.

- **a** Calculate the value of the investment after 2 years.
- **b** Complete the table below:

n	1	2	3	4	5	6	7
A							

- **c** Sketch the graph of the function.
- **d** How many years, n, does it take for the balance to reach $\$20\,988.08$?

29 $\$3000$ is invested at 18% p.a., with interest compounded annually.

- **a** Write the equation for the value, A, of the investment after n years.
- **b** Find the value of the investment after 6 years.
- **c** Complete the table below:

n	1	2	3	4	5	6	7
A							

- **d** Sketch the graph of A.

30 Vanessa invests $\$1000$ into a term deposit, [redacted] is compounded at a rate r each year, as shown in the following table:

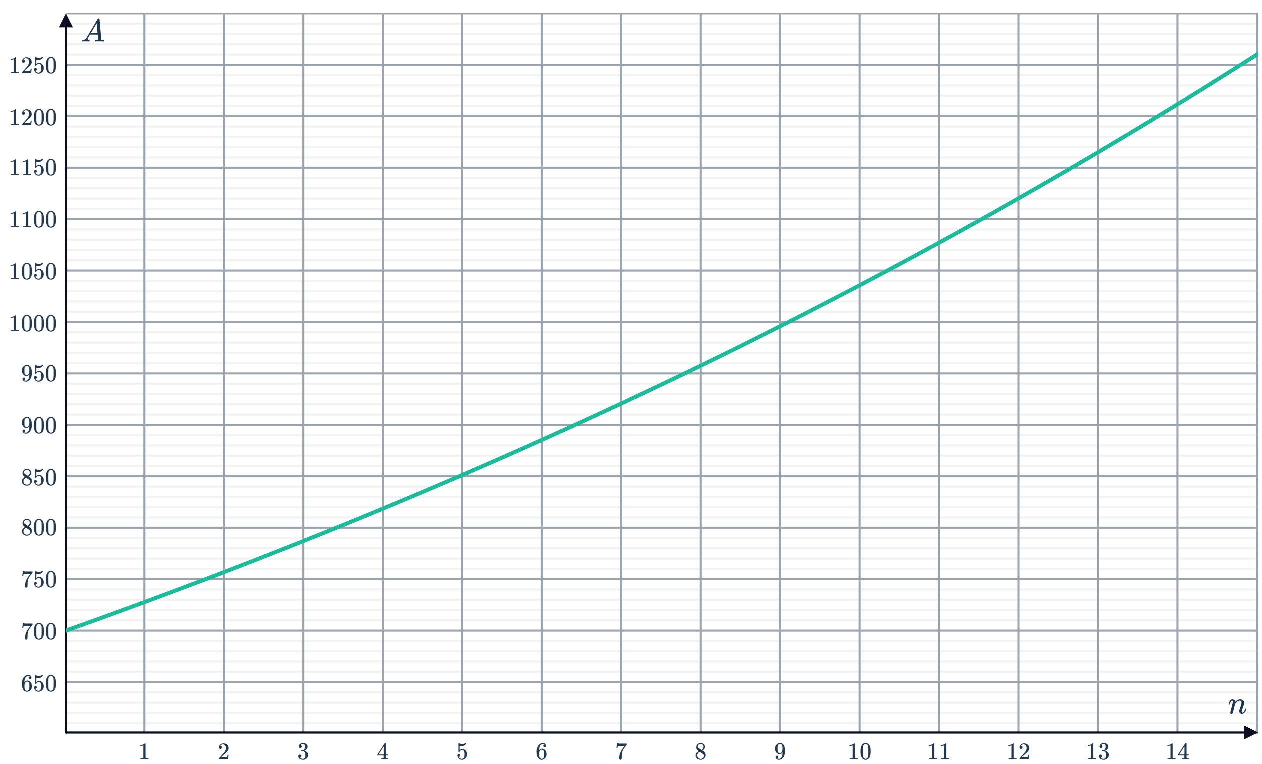
Years passed (n)	0	1	2
Value of investment $(\$A)$	1000	1020	1040.40

a Find the value of r from the table of values.

b Write a function, A, to model the value of the investment after n years.

c Find the value of the investment after 6 years, assuming the rate stays constant.

31 $\$700$ is invested at 4% p.a. compound interest. The value of the money, A, after n years is given by $A = 700\left(1+\dfrac{4}{100}\right)^n$. The function has been graphed below:

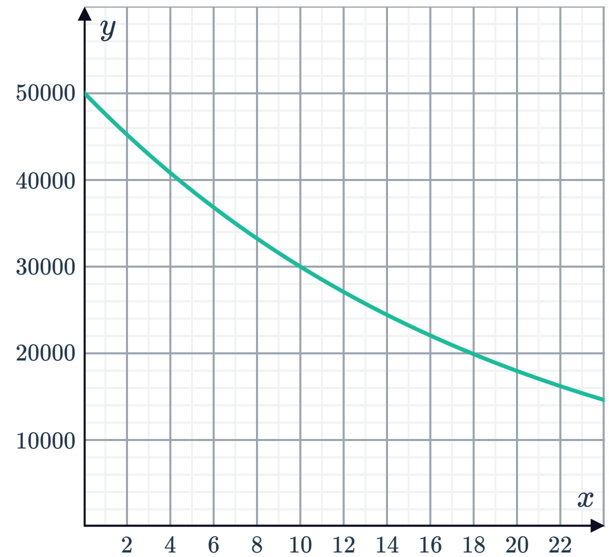


a Using the equation $A = 700\left(1+\dfrac{4}{100}\right)^n$, determine how much money would be in the account after 7 years, to the nearest dollar.

b According to the graph, approximately how many years does it take for the balance to reach $\$1120$?

c How much interest is earned during the 7th year?

32 Peter received a lump sum payment of $\$50$ [redacted] an insurance claim. Every month, he uses 5% of the remaining funds. The funds, y, after x months is shown in the graph:



a How much would he use in the first month?

b How much would be left after the first month?

c From the graph, estimate the amount left after one and a half years.

d From the graph, estimate for the number of months until the amount reaches $\$30\,000$.

33 A $\$110\,000$ investment earns interest at 4% p.a. compounded semi-annually over 5 years. The compound interest table shows the future value of a $\$1000$ investment compounded at various interest rates and over a certain number of periods.

Interest rate per period	10 years	20 years	30 years	40 years	50 years
0.5%	1051.14	1104.90	1161.40	1220.79	1283.23
1%	1104.62	1220.19	1347.85	1488.86	1644.63
1.5%	1160.54	1346.86	1563.08	1814.02	2105.24
2%	1218.99	1485.95	1811.36	2208.04	2691.59
2.5%	1280.08	1638.62	2097.57	2685.06	3437.11

Using the table, calculate:

a The future value of this investment.

b The total amount of interest earned.

34 Bank A pays interest at a rate of 5.3% p.a. compounded annually, while Bank B offers a rate of 5% p.a. with interest compounded quarterly.

a Form an equation for A, the amount $\$P$ would grow to after t years if invested with:

i Bank A.

ii Bank B.

b State the effective annual rate for Bank B as a percentage, to two decimal places.

c Which bank offers the better deal?

35 A $\$150\,000$ investment earns interest at 24% compounded monthly over 10 years. The following table shows the balance on a $\$1000$ investment for a particular number of periods, n:

Interest rate per period	$n = 115$	$n = 120$	$n = 125$	$n = 130$	$n = 135$
1%	3140.20	3300.39	3468.74	3645.68	3831.65
2%	9750.34	$10\,765.16$	$11\,885.61$	$13\,122.67$	$14\,488.49$
3%	$29\,942.00$	$34\,710.99$	$40\,239.55$	$46\,648.66$	54078.59
4%	$90\,956.56$	$110\,662.56$	$134\,637.93$	$163\,807.62$	$199\,297.02$
5%	$273\,381.67$	$348\,911.99$	$445\,309.93$	$568\,340.86$	$725\,362.96$

Using the table, calculate:

a The future value of this investment.

b The total amount of interest earned.

36 Maria has $\$1000$ to invest for 4 years and would like to know which of three investment plans to choose:

- Plan 1: invest at 4.98% p.a. interest, compounded monthly.
- Plan 2: invest at 6.44% p.a. interest, compounded quarterly.
- Plan 3: invest at 5.70% p.a. interest, compounded annually.

a Calculate the future value of each investment plan.

b State which plan Maria should choose for maximum return on her investment.

37 Grace wants to invest $\$1300$ at 2% p.a for 6 years. She has two investment options, compounding quarterly or compounding monthly. Calculate how much extra the investment is worth if it is compounded monthly rather than quarterly.